SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: OPERATING SYSTEMS **COURSE CODE: CSA04**

COURSE OUTCOME COVERED

CO2: Analyze process management and deadlock handling techniques to improve CPU utilization and system performance

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QUESTIONS: 5 **Total Marks: 100**

96/100

Annexure A

Analytical Problem Solving

Code of Conduct:

I **Galla Sai Siddartha(192425093)** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student : **Galla Sai Siddartha**

24/25

SAMPLE QUESTIONS

(QUESTIONS CAN BE CHANGED AS PER YOUR REQUIREMENTS)

Question 1: CPU Scheduling – Waiting Time & Turnaround Time Analysis

A set of processes arrive at time 0 with the following burst times:

Process	Burst Time (ms)
P1	10
P2	4
P3	6
P4	2

a. Calculate Waiting Time and Turnaround Time for each process using SJF (Non-preemptive).
b. Compute the average waiting time and average turnaround time.
c. Interpret whether SJF improves CPU utilization compared to FCFS for this workload.

(20 Marks) (BL4)

Question 2: Round Robin Scheduling – Time Quantum Impact

Consider the following processes:

Process	Burst Time (ms)
P1	20
P2	5
P3	13
P4	8

Let the time quantum = 4 ms.

a. Construct the Gantt chart for Round Robin.
b. Calculate Waiting Time and Turnaround Time of each process.
c. Analyze how changing the quantum to 10 ms would affect system performance (CPU utilization & context switching).

(20 Marks) (BL4)

Question 3: Deadlock – Safety Check Using Banker's Algorithm

A system has 3 resource types (A, B, C). Total resources:
A = 10, B = 5, C = 7.

The Allocation and Maximum Need matrices are:

Allocation Matrix

Process	A	B	C
P1	0	1	0
P2	2	0	0
P3	3	0	2
P4	2	1	1
P5	0	0	2

Maximum Matrix

Process	A	B	C
P1	7	5	3
P2	3	2	2
P3	9	0	2
P4	4	2	2
P5	5	3	3

a. Calculate the Need Matrix.
b. Determine if the system is in a safe state using Banker's Algorithm.
c. Provide a valid safe sequence if it exists.

(20 Marks) (BL4)

Question 4: Deadlock Detection – Resource Allocation Graph (RAG)

A system has the following resource allocations:

- P1 is holding R1 and waiting for R2
- P2 is holding R2 and waiting for R3

- P3 is holding R3 and waiting for R1

a. Draw the Resource Allocation Graph (describe in text).
b. Determine whether the system is in deadlock.
c. Recommend a strategy (prevention OR avoidance OR recovery) to resolve it and justify the method.

(20 Marks)(BL4)

Question 5: CPU Utilization – Multiprogramming Level Analysis

A system has the following process behavior:

- Each process spends 20% time in CPU and 80% waiting for I/O.
- Degree of multiprogramming = 4 processes

CPU Utilization formula:

$$U = 1 - (p^n)$$

where:

- p = probability a process is waiting for I/O,
- n = number of processes in memory

a. Calculate CPU Utilization.
b. Compute CPU utilization if the degree of multiprogramming increases to 6.
c. Interpret which configuration improves system performance and why.

(20 Marks)(BL4)

RUBRICS FOR CALCULATION

Analytical Problems

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

TOTAL MARKS AWARDED

SIGNATURE

FACULTY

CPU Scheduling - SJF (Non-preemptive)

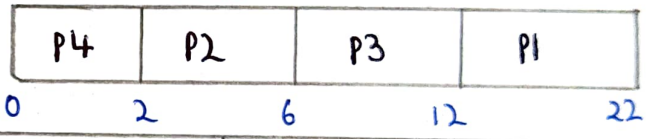
Given

Process	Burst time (ms)
P1	10
P2	4
P3	6
P4	2

All processes arrive at time 0.

(a) Calculate Waiting time and turnaround time

Execution Order : P4 → P2 → P3 → P1



Process	Arrival time	Burst time	Completion time	Waiting time	Turnaround time
P1	0	10	22	12	22
P2	0	4	6	2	6
P3	0	6	12	6	12
P4	0	2	2	0	2

(b) Average Waiting time and Turnaround time

Average WT = $\frac{12+2+6+0}{4}$ = 5ms Average Turnaround Time = $\frac{42}{4}$ = 10.5 ms

(c) Compare SJF with FCFS

Under FCFS, the execution order is P1 → P2 → P3 → P4

- FCFS average waiting time = (0+10+14+20)/4 = 11 ms
- FCFS average turnaround time = (10+14+20+22)/4 = 16.5 ms
- SJF average waiting time = 5 ms
- SJF average turnaround time = 10.5 ms.

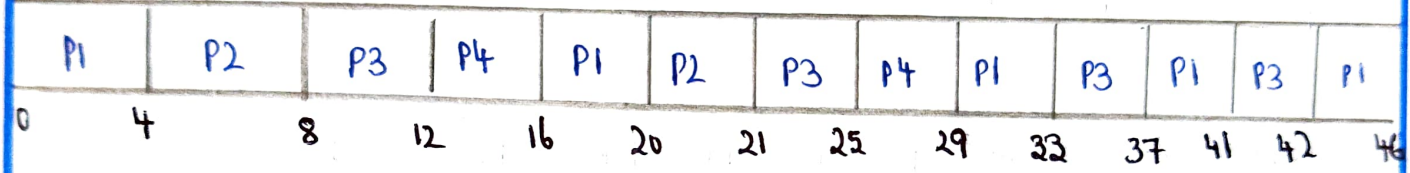
2) Round Robin Scheduling - Time Quantum Impact

Given,

Process	Burst Time (ms)
P1	20
P2	5
P3	13
P4	8

(a) Construct the Gantt Chart

Time Quantum = 4 ms



(b) Calculate waiting time and turnaround time

Turnaround Time = Completion Time - Arrival Time

Waiting Time = Turnaround Time - Burst Time.

Process	Burst Time	Completion time	Turnaround time	Waiting Time
P1	20	46	46	26
P2	5	21	21	16
P3	13	42	42	29
P4	8	29	29	21

Average Waiting Time : $\frac{26+16+29+31}{4} = 23$ ms Average turnaround time = 34.5 ms

(c) Effect of increasing the quantum to 10 ms

Factor	Quantum = 4 ms	Quantum = 10 ms
Context Switching	More frequent	Less frequent
Response Time	Better for interactive tasks	May be slower for some process
Scheduling behaviour	Higher	Lower.

Deadlock - Safety Check Using Banker's Algorithm

Given total resource:

A = 10 B = 5 C = 7

(a) Calculate the Need Matrix

Formula

Need = Maximum - Allocation

Process	Allocation (A,B,C)	Maximum (A,B,C)	Need (A,B,C)
P1	(0,1,0)	(7,5,3)	(7,4,3)
P2	(2,0,0)	(3,2,2)	(1,2,2)
P3	(3,0,2)	(9,0,2)	(6,0,0)
P4	(2,1,1)	(4,2,2)	(2,1,1)
P5	(0,0,2)	(5,3,3)	(5,3,1)

(b) Determine whether the system is in a safe state:

First, calculate the available resources.

Total allocated resource:

A = 0+2+3+2+0 = 7 B = 1+0+0+1+0 = 2 C = 0+0+2+1+2 = 5

Available = (3,3,2)

Step	Process Selected	Need	Work before	Work after Completion
1	P2	(1,2,2)	(3,3,2)	(5,3,2)
2	P4	(2,1,1)	(5,3,2)	(7,4,3)
3	P1	(7,4,3)	(7,4,3)	(7,5,3)
4	P3	(6,0,0)	(7,5,3)	(10,5,5)
5	P5	(5,3,1)	(10,5,5)	(10,5,7)

(c) Valid safe Sequence

P2 → P4 → P1 → P3 → P5

The system is in a safe state because all five processes can completed in the sequence without causing deadlock.

4) Deadlock Detection - Resource Allocation Graph

Given

- P1 is holding R1 and waiting for R2
- P2 is holding R2 and waiting for R3
- P3 is holding R3 and waiting for R1

(a) Draw (or) describe the Resource Allocation Graph

A Resource Allocation Graph contains:

Process nodes :- P1, P2, P3 are represented by circles

Resource nodes :- R1, R2, R3 represented by square

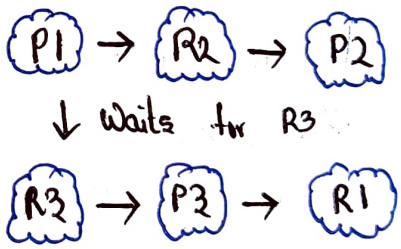
Allocation Edge :- Resource → Process

Request Edge :- Process → Resource.

The graph has these edge :-

Edge	Meaning
R1 → P1	R1 is allocated to P1
P1 → R2	P1 requests R2
R2 → P2	R2 is allocated to P2
P2 → R3	P2 requests R3
R3 → P3	R3 is allocated to P3
P3 → R1	P3 requests R1

Cycle Representation



The cycle closes because P3 request R1, which is held by P1

(b) Determine whether the system is in deadlock

Yes, the system is in deadlock, assuming each resource has a single instance.

The circular wait is.

P1 → R2 → P2 → R3 → P3 → R1 → P1

4) Recommend a strategy to resolve the deadlock

Recommended Strategy :- Deadlock recovery using resource preemption

Steps:-

1. Identify a process involved in the circular wait.
2. Select a suitable victim process, considering its priority and the cost of restarting it.
3. Preempt its resource, such as R1, if the resource can safely be taken back
5. Restart or roll back the victim process when resource becomes available.

5) CPU Utilization - Multiprogramming Level Analysis

Given

Each process spends 20% of its time executing on the CPU

Each process spends 80% of its time waiting for I/O

Degree of multiprogramming, $n = 4$ processes initially.

Probability of a process waiting for I/O, $p = 0.8$

(a) CPU Utilization for 4 processes

Substitute $p = 0.8$ and $n = 4$

$$U = 1 - p^n$$

$$U = 1 - (0.8)^4$$

$$= 0.5904$$

CPU Utilization = 59.04%

(b) Calculate CPU Utilization for 6 processes

$p = 0.8 \quad n = 6$

$$U = 1 - p^n$$

$$U = 1 - (0.8)^6$$

$$= 0.7379$$

CPU Utilization = 73.79%

(c) Interpret which configuration improves system performance

4 process → 59.04%

6 process → 73.79%

The configuration with 6 processes provide better CPU Utilization than the configuration with 4 process. Therefore, increasing the degree of multiprogramming reduce CPU idle time and improve overall performance.